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A TIGHT LINEARIZATION AND AN ALGORITHM FOR ZERO-ONE QUADRATIC PROGRAMMING PROBLEMS*

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This paper is concerned with the solution of linearly constrained zero-one quadratic programming problems. Problems of this kind arise in numerous economic, location decision, and strategic planning situations, including capital budgeting, facility location, quadratic assignment, media selection, and dynamic set covering. A new linearization technique is presented for this problem which is demonstrated to yield a tighter continuous or linear programming relaxation than is available through other methods. An implicit enumeration algorithm which uses Lagrangian relaxation, Benders' cutting planes, and local explorations is designed to exploit the strength of this linearization. Computational experience is provided to demonstrate the usefulness of the proposed linearization and algorithm.
(ZERO-ONE QUADRATIC PROGRAMMING; LINEARIZATION TECHNIQUES; IMPLICIT ENUMERATION; LAGRANGIAN RELAXATION; BENDERS' DECOMPOSITION)

1. Introduction

This paper is concerned with linearly-constrained zero-one quadratic programming problems (QP). Problems of this structure arise in numerous economic, facility location, and strategic planning situations. These include, for example, the media selection problem described by Zangwill (1965), the capital budgeting problem considered by Laughhunn (1970) and by Peterson and Laughhunn (1971), certain facility location problems given in Vaish (1974), a multitude of applications associated with quadratic assignment problems (see Sherali 1979), and dynamic facility relocation problems such as the dynamic set covering problem formulated in Chrissis *et al.* (1978).

Basically, the available solution procedures for Problem QP may be classified as attempting either to solve the problem directly or to transform Problem QP into an equivalent linear mixed-integer program, and then solve the latter problem. The direct methods involve partitioning techniques or implicit enumeration strategies. Lazimy (1982) considers a more general version of Problem QP, i.e., a mixed-integer quadratic programming problem, and demonstrates that one can transform this problem into an equivalent program without adding new variables or constraints. This renders the problem more amenable to Geoffrion's (1972) generalized Benders' technique, in that the cutting planes derived are linear in the integer variables. Balas (1969) suggests an alternative partitioning algorithm for the same class of problems, but requires the addition of an extra set of variables. Both these methods, however, are based on the use of a Dorn (1960) type of duality, and therefore require appropriate convexity assumptions in regard to the objective function. Another direct-search cutting plane algorithm is suggested by Kunzi and Oettli (1963) for all-integer quadratic programming problems.

The direct implicit enumeration procedures developed for Problem QP are more straightforward. These methods include the works of Cabot and Francis (1970), Hansen (1972), Mao and Wallingford (1968), and McBride and Yormark (1980). Recently, Carter (1984) has suggested methods for improving various direct-search schemes by

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strategically transforming indefinite zero-one quadratic problems into equivalent (in the integer sense) positive definite quadratic problems.

Another approach for Problem QP, as well as for more general zero-one polynomial programming problems, is to transform the nonlinear integer program into an equivalent linear mixed-integer program through the use of additional variables and constraints. Fortet (1959, 1960) (see also Watters 1967 and Zangwill 1965) has proposed a linearization technique for zero-one polynomial programming problems which replaces each polynomial term with a single additional zero-one variable and two additional constraints. More concise zero-one linear programming formulations of zero-one polynomial programming problems are found in Glover and Woolsey (1973). These formulations, as well as the ones by Fortet, require the adoption of additional zero-one variables. On the other hand, some proposed linearization techniques do not introduce additional zero-one variables, but introduce only additional continuous variables and additional constraints. The linearizations due to Glover (1975), Glover and Woolsey (1974), and Rhys (1970) are examples of such techniques. A similar technique for the special case of quadratic assignment problems has been presented by Bazaraa and Sherali (1980).

The experience reported by Bazaraa and Sherali (1980) and a comment by McBride and Yormark (1980) seem to indicate that the linearization techniques do not yield very effective solution procedures for Problem QP. In this paper, we demonstrate why this may be the case and suggest a method to dramatically improve upon the performance of algorithms based on equivalent linearizations. More specifically, we introduce in §2 a new linearization technique for Problem QP. The resulting linearization is shown to theoretically dominate the linearizations found in the literature. In §3 we propose an effective algorithm that exploits the strength of the derived linearization. This algorithm uses Lagrangian relaxation imbedded within an implicit enumeration framework, and generates Benders' cuts and conducts local explorations in order to enhance its performance. In the final section of this paper we provide computational experience. In particular, the strength of the linearization is verified by exhibiting the dramatic savings in computational effort that is achieved in comparison with the implementation of the same algorithmic strategies on an alternative existing linearization of Problem QP.

2. A Linearization Technique

The zero-one quadratic programming problem may be stated mathematically as follows:

$$\begin{aligned} &\text{minimize} && \sum_{i=1}^{m} c_i x_i + \sum_{i=1}^{m-1} \sum_{j=i+1}^{m} D_{ij} x_i x_j \\ &\text{subject to} && x \in X, \qquad x \text{ binary} \end{aligned} \tag{QP}$$

where $X \equiv X_1 \cap X_2 \cap X_3$ is a nonempty polyhedron with
$X_1 = \{x \in R^m\colon \sum_{i=1}^m a_{ki} x_i = b_k \text{ for } k = 1, \ldots, K\}$,
$X_2 = \{x \in R^m\colon \sum_{i=1}^m G_{li} x_i \geqslant g_l \text{ for } l = 1, \ldots, L\}$, and
$X_3 = \{x \in R^m\colon 0 \leqslant x_i \leqslant 1 \text{ for } i = 1, \ldots, m\}$.

In this section, we present a linearization technique which transforms Problem QP into an equivalent zero-one mixed-integer linear programming problem. The emphasis in deriving this linearization is on achieving as tight a linear programming relaxation as possible, without regard to the size of the resulting equivalent linear mixed-integer program. In the following section, we demonstrate how to design algorithmic strategies that exploit the tightness of this representation of the problem, without being slowed down by its size. The motivation for providing an equivalent representation with a

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tight linear programming relaxation is borne out by several studies in the discrete optimization literature (see Adams and Sherali 1983, Magnanti and Wong 1981, McDaniel and Devine 1977, and Rardin and Unger 1976, for example).

Toward this end, consider the following sets of operations:

(i) form mK constraints by multiplying the K equalities of X_1 by each x_j, $j = 1, \ldots, m$;

(ii) form mL constraints by multiplying the L inequalities in X_2 by each x_j, $j = 1, \ldots, m$;

(iii) form mL constraints by multiplying the L inequalities in X_2 by each $(1 - x_j)$, $j = 1, \ldots, m$; and

(iv) form $3m(m-1)/2$ constraints by multiplying, for each i, the $x_i \geqslant 0$ inequality in X_3 by each $(1 - x_j)$, $j = 1, \ldots, m$, $j \neq i$, and by multiplying, for each $i = 1, \ldots, m-1$, the $x_i \leqslant 1$ inequality in X_3 by each $(1 - x_j)$, $j = i+1, \ldots, m$.

Then substitute $w_{ij} \equiv x_i x_j \; \forall \, (i, j)$, $w_{ij} \geqslant 0$. Noting that one need only define w_{ij} for $i = 1, \ldots, m-1$, $j = i+1, \ldots, m$, and recognizing that x_i^2 may equivalently be written as x_i for $i = 1, \ldots, m$, we obtain the following linear mixed-integer representation of Problem QP. (Henceforth, wherever obvious, the limits on index values are suppressed for notational ease.)

$$\text{minimize} \quad z = \sum_i c_i x_i + \sum_{i<j}\sum D_{ij} w_{ij} \tag{MIP}$$

$$\text{subject to} \quad (a_{kj} - b_k)x_j + \sum_{i<j} a_{ki} w_{ij} + \sum_{i>j} a_{ki} w_{ji} = 0 \qquad \forall \, (j,k), \tag{2.1}$$

$$\sum_i G_{li} x_i - g_l \geqslant \sum_{i<j} G_{li} w_{ij} + \sum_{i>j} G_{li} w_{ji} + (G_{lj} - g_l) x_j \geqslant 0 \qquad \forall \, (j,l), \tag{2.2}$$

$$x_i - w_{ij} \geqslant 0 \qquad \forall \, (i,j), \quad i<j, \tag{2.3}$$

$$x_j - w_{ij} \geqslant 0 \qquad \forall \, (i,j), \quad i<j, \tag{2.4}$$

$$-x_i - x_j + w_{ij} \geqslant -1 \qquad \forall \, (i,j), \quad i<j, \tag{2.5}$$

$$w_{ij} \geqslant 0 \qquad \forall \, (i,j), \quad i<j, \tag{2.6}$$

$$x \in X_1, \quad x \in X_3, \quad x \text{ binary}. \tag{2.7}$$

Several remarks are in order at this point. First, since the constraints in X_1 are equalities, we need not generate another set of constraints by multiplying these constraints by each $(1 - x_j)$, $j = 1, \ldots, m$. This would only duplicate (2.1), given that $x \in X_1$. Second, (2.2) implies the inequalities in X_2, and so $x \in X_2$ is omitted in (2.7). Third, one ought to preprocess QP and perform standard logical tests in order to possibly restrict the values of some binary variables to zero or one at the outset. These tests could include verifying whether or not the minimum and maximum values which each variable x_j can take over the region X are indeed zero and one respectively. Fourth, depending on the specific problem structure, some constraints in MIP may be omitted without affecting the continuous relaxation. For example, given that X_3 (or some subset thereof) is implied by X_1 and X_2, as is the case when X_1 consists of set partitioning constraints, Problem MIP need not include X_3 (or the corresponding subset) and need not include the associated constraints (2.3), (2.4), and (2.5). Finally, cross product terms between integer and some other continuous variables (which are included in the constraint set as well) may be readily accommodated. Adams and Sherali (1983) demonstrate this for a discrete version of the Bilinear Programming Problem (see Konno 1976). One can handle quadratic terms in the binary variables which appear in the constraints in a similar manner.

Let us first demonstrate the equivalence between Problems QP and MIP.

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THEOREM 1. *Problems QP and MIP are equivalent in the following sense. Given any (x) feasible to QP, there exists a w such that (x, w) is feasible in MIP with the same objective value. Conversely, given any (x, w) feasible in MIP, the corresponding solution (x) is feasible in QP with the same objective value.*

PROOF. Given any (x) feasible in QP, compute $w_{ij} = x_i x_j \ \forall\ (i, j)$ with $i < j$. Then (x, w) is feasible in MIP with the same objective value. Conversely, let (x, w) be feasible in MIP. It is sufficient to show that $w_{ij} = x_i x_j \ \forall\ (i, j)$, $i < j$ because then x is feasible in QP with the same objective value. Consider any (i, j) with $i < j$. If x_i or x_j equals zero, (2.3), (2.4), and (2.6) imply that $w_{ij} = 0$. On the other hand, if $x_i = x_j = 1$, then (2.3), (2.4), and (2.5) imply that $w_{ij} = 1$. This completes the proof. □

Observe that the proof of Theorem 1 indicates that the program MIP1 obtained from MIP by deleting constraints (2.1) and (2.2), but explicitly adding $x \in X_2$ in (2.7), is, in fact, equivalent to QP. Indeed, MIP1 is one linearization proposed by Glover and Woolsey (1974) for Problem QP. The constraints (2.1) and (2.2), other than enforcing $x \in X_2$, are inconsequential in the integer programming sense but, as will be shown, do provide a tighter linear programming relaxation for QP.

Consider now an alternative linearization technique for QP which also does not employ additional zero-one variables, namely, that due to Glover (1975). This technique produces the following equivalent linear mixed-integer representation MIP2 of Problem QP

$$\text{minimize} \qquad z = \sum_i c_i x_i + \frac{1}{2}\sum_i B_i \tag{MIP2}$$

subject to

$$\sum_{j<i} D_{ji}x_j + \sum_{j>i} D_{ij}x_j - d_i^+(1-x_i) \leq B_i \qquad \forall\, i,$$

$$\sum_{j<i} D_{ji}x_j + \sum_{j>i} D_{ij}x_j - d_i^-(1-x_i) \geq B_i \qquad \forall\, i, \tag{2.8}$$

$$d_i^- x_i \leq B_i \leq d_i^+ x_i \qquad \forall\, i, \tag{2.9}$$

$$x \in X, \quad x \text{ binary}, \tag{2.10}$$

where d_i^- and d_i^+ are, respectively, the minimum and maximum values of the expression $\sum_{j<i} D_{ji}x_j + \sum_{j>i} D_{ij}x_j$ over the set X, for each i. As suggested by Glover (1975), one may equivalently use simple lower and upper bounds on the optimal values of these problems for d_i^- and d_i^+ respectively, although this of course weakens the linear programming relaxation. The next theorem and example show that MIP provides a tighter continuous relaxation of Problem QP than does MIP2. (This is clearly the case with respect to MIP1.)

THEOREM 2. *Let $\overline{MIP}$ and $\overline{MIP2}$, respectively, denote the continuous relaxations of Problems MIP and MIP2 obtained by omitting the binary restrictions on the x-variables. Then $\overline{MIP}$ is a tighter relaxation of QP than is $\overline{MIP2}$ in the sense that for any $(\bar{x}, \bar{w})$ feasible to $\overline{MIP}$, there exists a vector $\bar{B} \equiv (\bar{B}_1, \ldots, \bar{B}_m)$ such that $(\bar{x}, \bar{B})$ is feasible to $\overline{MIP2}$ with the same objective value.*

PROOF. For any scalar $\psi \geq 0$, let the set $Z(\psi) \equiv \{z \geq 0\colon \sum_j a_{kj}z_j = \psi b_k \ \forall\ k,\ \sum_j G_{lj}z_j \geq \psi g_l \ \forall\ l,\ z_j \leq \psi \ \forall\ j\}$ be obtained by multiplying the right-hand sides of the constraints of X by ψ. Then, noting the above definitions of d_i^- and $d_i^+ \ \forall\ i$, it follows from linear programming duality that for any $\psi \geq 0$, $z \in Z(\psi)$ implies

$$\psi d_i^- \leq \sum_{j<i} D_{ji}z_j + \sum_{j>i} D_{ij}z_j \leq \psi d_i^+ \qquad \forall\, i. \tag{2.11}$$

Now, let $(\bar{x}, \bar{w})$ be any feasible solution to MIP. Since $\bar{x} \in X$ from (2.2) and (2.7), it is sufficient to show that by defining

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$$\bar{B}_i \equiv \sum_{j<i} D_{ji}\bar{w}_{ji} + \sum_{j>i} D_{ij}\bar{w}_{ij} \qquad \forall\, i, \tag{2.12}$$

we obtain that $(\bar{x}, \bar{B})$ is feasible to (2.8) and (2.9), since $(\bar{x}, \bar{B})$ yields the same objective value in $\overline{\text{MIP2}}$ as $(\bar{x}, \bar{w})$ does in $\overline{\text{MIP}}$.

Toward this end, for any i, identify $\bar{z}_j = \bar{w}_{ij}\ \forall\, j > i$, $\bar{z}_j = \bar{w}_{ji}\ \forall\, j < i$, $\bar{z}_i = \bar{x}_i$, and $\psi = \bar{x}_i$. Then from (2.1), the right-hand inequalities in (2.2), and from (2.3), (2.4), and (2.6), along with the fact that $\bar{x}_i \geqslant 0$, it follows that $\bar{z} \in Z(\bar{x}_i)$ and hence from (2.11) and (2.12), we obtain that (2.9) holds for each i. Now for any i, identify $\bar{z}_j = (\bar{x}_j - \bar{w}_{ij})\ \forall\, j > i$, $\bar{z}_j = (\bar{x}_j - \bar{w}_{ji})\ \forall j < i$, $\bar{z}_i = 0$, and $\psi = (1 - \bar{x}_i)$. Observing from (2.1) and $\bar{x} \in X_1$ that

$$\sum_{j<i} a_{kj}(\bar{x}_j - \bar{w}_{ji}) + \sum_{j>i} a_{kj}(\bar{x}_j - \bar{w}_{ij}) = b_k(1 - \bar{x}_i) \qquad \forall\, k, \qquad \text{for each } i,$$

and noting the left-hand inequalities in (2.2), and the inequalities (2.3), (2.4), and (2.5), along with the fact that $(1 - \bar{x}_i) \geqslant 0$, it follows that $\bar{z} \in Z(1 - \bar{x}_i)$. Hence, from (2.11) and (2.12), we obtain that (2.8) holds for each i. This completes the proof. □

To demonstrate that the converse of Theorem 2 is not true in general, and therefore that $\overline{\text{MIP}}$ can provide a strictly tighter relaxation of Problem QP than does $\overline{\text{MIP2}}$, consider the following example. Define $X = \{(x_1, x_2, x_3, x_4)\colon x_1 + x_2 + x_4 \geqslant 1,\ x_2 + x_3 \geqslant 1,\ x_1 + x_3 + x_4 \geqslant 1,\ 0 \leqslant x \leqslant 1\}$, and let the cost coefficients be given by $c_i = 0\ \forall\ i$ and $(D_{12}, D_{13}, D_{14}, D_{23}, D_{24}, D_{34}) = (2, 4, 6, 3, 6, 6)$. Then the optimal solution to Problem QP is $x = (1, 1, 0, 0)$ with the optimal objective value equal to 2. It is readily verified that Problem $\overline{\text{MIP1}}$, Problem MIP1 with the x binary restrictions relaxed, is solved by $x = (\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, 0)$ and $w_{ij} = 0\ \forall\ (i, j)$, $i < j$, with optimal objective value 0. Similarly, $\overline{\text{MIP2}}$ is also solved by $x = (\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, 0)$, but with $B_1 = 1$, $B_2 = 1$, $B_3 = 1.5$, and $B_4 = 0$ with an optimal objective value of 1.75. However, the optimal solution to $\overline{\text{MIP}}$ is $x = (1, 1, 0, 0)$, $w_{12} = 1$, $w_{ij} = 0$ otherwise, with objective value 2, and hence is verified to be optimal to Problem QP.

Note that the recommendation due to Glover and Woolsey (1974) to replace (2.3) and (2.4) in Problem MIP1 by the surrogates $\sum_{j>i} w_{ij} + \sum_{j<i} w_{ji} \leqslant (m - 1)x_i\ \forall\ i$ further weakens the linearization in the continuous sense, though it maintains an equivalent mixed-integer representation. Furthermore, the strategy of Peterson (1971), when applied to Problem QP, results in the same formulation as MIP1 and hence is also dominated by Problem MIP. The method suggested by Rhys (1970) for the case of unconstrained zero-one quadratic programming problems essentially results in Problem MIP1 with, of course, the constraint sets X_1 and X_2 absent. Finally, it can be shown that the bound obtained for Problem QP via an extension of Hammer *et al.*'s (1984) theory of roof duality to the linearly-constrained case is precisely the optimal objective function value to Problem $\overline{\text{MIP1}}$.

Constraints (2.1) and (2.2) of Problem $\overline{\text{MIP}}$ tend to restrict, beyond that of available linearizations, the range of values w may take on for any given (nonbinary) $x \in X$. Of course, the relative strength of these constraints is somewhat dependent on the set X. If, for example, the set X is composed of set partitioning constraints, then one might expect the linearization to be significantly stronger than when the set X is composed of set covering constraints. In both these instances, however, the density (proportion of nonzero coefficients) of the constraint set may affect the strength of the linearization, becoming generally stronger with a lower density. As a result, one may wish to choose judiciously from amongst the available constraints in (2.1) and (2.2) in constructing a linearization of the type MIP for some specific problem, depending on the nature of these constraints.

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3. Proposed Algorithm

In this section, we first provide a general overview of the proposed algorithm and then present specific details for some important algorithmic steps in later subsections.

Basically, the algorithm is an implicit enumeration strategy that uses a combination of the breadth- and depth-first methods. In order to curtail storage, and still provide the benefits of a breadth-first approach, the binary enumeration search tree over the x-variables is developed in a breadth-first fashion whenever the number of active tree-end nodes is less than a specified maximum, and proceeds with a selected node in a depth-first fashion otherwise.

At node zero (the beginning), we perform logical tests as in Salkin (1975) to probe and possibly a priori restrict some x-variables at binary values. Thereafter, we determine $\bar{\nu}_l = \min\{\sum_i x_i : x \in X\}$ and $\bar{\nu}_u = \max\{\sum_i x_i : x \in X\}$ and generate the constraints

$$\nu_l \leqslant \sum_i x_i \leqslant \nu_u \tag{3.1}$$

where ν_l and ν_u are respectively the rounded-up and rounded-down values of $\bar{\nu}_l$ and $\bar{\nu}_u$. Constraints (3.1) are added to X and considered a part thereof henceforth; they serve not only to potentially further tighten the continuous relaxation, but they are also used in the sequel to compute knapsack-based lower bounds.

Next, we consider a Lagrangian dual of $\overline{\text{MIP}}$ and solve this problem to near optimality using a subgradient optimization method. The motivation here is to determine quickly a good quality dual feasible solution with which to generate a Benders' cut in the x-variables. (The direct solution of the linear program $\overline{\text{MIP}}$ may be too time consuming because of its size.) For a near-optimal dual solution, this cut is a close approximation to the strongest surrogate Benders' cut (see Rardin and Unger 1976). Since the lower bound available from this cut is close to that available from the linear programming relaxation $\overline{\text{MIP}}$ (the two are equal if the cut is generated from a dual-optimal solution), this constraint provides tight lower bounds, assuming that $\overline{\text{MIP}}$ is a tight continuous relaxation of MIP. As a means of strengthening the derived bounds, we suitably update this constraint at each node within the implicit enumeration framework, depending on the variables that are fixed at binary values. Of course, both in deriving and in updating this cut, one must strike a judicious compromise between the strength of the cut (and its ensuing benefits) and the overall effort expended. §3.1 and §3.2, as well as the computational results in §4, provide specific guidelines. Let us denote this cut (at node 0) as

$$z \geqslant \delta^0 + \sum_i \gamma_i^0 x_i. \tag{3.2}$$

Upon deriving (3.2), we record the lower bound

$$\nu(\overline{\text{MIP}}) \equiv \min\{\delta^0 + \sum_i \gamma_i^0 x_i : x \in X\} \tag{3.3}$$

on the optimal objective value of MIP and use the solution to (3.3) in order to possibly obtain an incumbent solution to MIP by truncating or rounding this solution in the spirit of Balas and Martin (1980). Of course, if X possesses a special structure such as a network structure or a set covering structure, then an incumbent is more readily available through (3.3).

Now the algorithm enters into the implicit enumeration phase. Given a certain node N being explored with I^+, I^-, and I^f respectively denoting the index sets of x-variables fixed at one, fixed at zero, and currently free, an attempt is made either to fathom the node or to select a branching variable to further partition and explore the reduced-node problem. This search process includes conducting logical tests, deriving one or

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more lower bounds as deemed judicious via the updated versions of (3.2) and other generated Benders' cuts, and conducting local explorations in order to detect improved incumbent solutions.

Specifically, after creating node N by branching on some designated variable, logical tests are performed in an attempt to further restrict some x-variables at binary values. During the entire node exploration process, whenever $|I^f| \leqslant 2$, the remaining possible completions are explicitly enumerated and the node is fathomed. Thereafter, an updated version of (3.2) at node N, namely,

$$z \geqslant \delta^N + \sum_{i \in I^f} \gamma_i^N x_i \tag{3.4}$$

is generated in the manner described in §3.2. Using (3.4), a trivial lower bound LB1 is first computed via the following bounded-interval knapsack linear program

$$\text{LB1} = \min \{\delta^N + \sum_{i \in I^f} \gamma_i^N x_i : \nu_l \leqslant \sum_{i \in I^f} x_i + |I^+| \leqslant \nu_u, 0 \leqslant x_i \leqslant 1 \ \forall\, i \in I^f\}. \tag{3.5}$$

If ν^* denotes the current incumbent objective value and if $(\epsilon + \text{LB1}) \geqslant \nu^*$, where $\epsilon > 0$ is some suitable tolerance, then the node is fathomed. Otherwise, denoting $\hat{x}_i$, for $i \in I^f$, as the optimal (binary) solution to (3.5), a binary vector $\tilde{x}$ is obtained by fixing $\tilde{x}_i = 1$ for $i \in I^+$, $\tilde{x}_i = 0$ for $i \in I^-$, and $\tilde{x}_i = \hat{x}_i$ for $i \in I^f$, and a Benders' cut

$$z \geqslant \Delta^N + \sum_{i \in I^f} \Gamma_i^N x_i \tag{3.6}$$

is generated based on $\tilde{x}$ as described in §3.3 below. Further, in case $\tilde{x}$ is feasible to X, the incumbent solution and value are updated if the objective value of $\tilde{x}$ in Problem QP, say $\tilde{z}$ (which may be obtained by substituting $x_i = \tilde{x}_i$, $\forall\ i \in I^f$, in (3.6)), is less than ν^*. (The current node is fathomed if $\tilde{x}$ is feasible to X and $\tilde{z} = \text{LB1}$.) Failing this, a strengthened lower bound LB2 is computed via the following problem:

$$\text{LB2} = \min \{z : (3.4), (3.6), \text{and the constraints in (3.5)}\}. \tag{3.7}$$

The motivation for (3.7) is that the constraints (3.4) and (3.6) complement each other well in that (3.6) lends strength to (3.4) in the vicinity of the solution $\tilde{x}$ at which (3.4) is the weakest, subject to the constraints in (3.5). Note that LB2 may be easily computed by solving the Lagrangian dual of (3.7) obtained by dualizing (3.4) and (3.6). Since the dual variables to these constraints are nonnegative and must sum to one, the problem reduces to maximizing a concave univariate function over the constraints in (3.5). We used the tangential approximation method (see Bazaraa and Shetty 1979) to solve this problem, exploiting the fact that, since it is a univariate problem, only the two currently binding constraints in the outer linearization need to be maintained. Generally, convergence was obtained in about four iterations.

If LB2 fails to fathom node N, then the cuts (3.4) and (3.6) are surrogated using the optimal dual multipliers available through (3.7) to yield the cut, say,

$$z \geqslant p^N + \sum_{i \in I^f} q_i^N x_i. \tag{3.8}$$

In case the optimal dual variable associated with (3.4) in the solution to (3.7) is not unity (or else (3.7) and (3.8) would essentially be (3.5) and (3.4) respectively), problem (3.5) is solved with its objective function replaced by the expression in (3.8). Denote by $\bar{x}$ the binary solution obtained by completing the current partial solution with the optimal (zero-one) solution to this problem. This solution is available when solving the Lagrangian dual subproblem to (3.7) at the optimal dual solution value. The solution $\bar{x}$ is then used to possibly update the current incumbent if it is feasible and improving.

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If node N is still not fathomed, then, using (3.8) along with the constraints of X, logical tests are performed to check if any further x-variables can be fixed at zero or one. If so, the partial solution is accordingly incremented and all the above operations at node N are repeated. Otherwise, a check is made to determine if it would be worthwhile to compute a somewhat more expensive lower bound. That is, if $\bar{x}$ is infeasible in X, and if LB2 and v^* are close enough in the sense that LB2 + 0.1 $(p^N + \sum_{i \in I^f} q_i^N -$ LB2$) \geqslant v^*$, then the linear program

$$\text{LB3} = \min \{p^N + \sum_{i \in I^f} q_i^N x_i : x \in X, x_i = 1 \text{ for } i \in I^+, x_i = 0 \text{ for } i \in I^-\} \tag{3.9}$$

is solved to compute a tighter lower bound, LB3. Let x' denote an optimal solution to (3.9). If no fathoming results from the use of LB3, then that index $i \in I^f$ for which x_i' is a maximum, with ties broken by selecting that index with the smallest value of q_i^N, is chosen as the branching variable index to be placed in I^+ upon the next selection of the active node N. Furthermore, the optimal solution x' to (3.9) may be rounded or truncated in order to possibly find an improved incumbent solution. If, however, LB3 is not computed, then a branching variable is chosen as above by using $\bar{x}$ in lieu of x'.

We found another feature to be useful in the foregoing schema. Whenever a new incumbent solution is discovered, we check the extreme points of the unit hypercube adjacent to this solution in order to possibly determine a further improving, feasible binary solution. As reported in §4, these local explorations helped in detecting further improving solutions on several occasions.

The following three subsections discuss details of the above algorithm: §3.1 describes the manner in which a strongest surrogate Benders' cut (3.2) is generated; §3.2 explains the procedure used for updating, based on the given active node, the dual variables for constraints (2.1) through (2.5), and hence updating the cut (3.2); and §3.3 presents in closed form an optimal dual solution to Problem QP for a fixed binary $x \in X$ with which to generate the Benders' cut (3.6).

3.1. *Generating the Strongest Surrogate Constraint*

The strongest surrogate constraint (see Rardin and Unger 1976) is essentially a Benders' cut, in terms of the "complicating" x-variables, derived from a dual optimal solution to Problem $\overline{\text{MIP}}$. Since $\overline{\text{MIP}}$ is a large linear program, one must compromise between the strength of the derived cut and the effort expended in obtaining it. We found that the following technique yields a good quality dual feasible solution within reasonable computational times. This solution provides an approximate strongest surrogate Benders' constraint, which is nonetheless a valid cut.

The technique adopted is basically a subgradient optimization procedure on a Lagrangian dual of $\overline{\text{MIP}}$. Letting $u = (u_{jk})$, $\alpha = (\alpha_{jl})$, $v = (v_{jl})$, $\beta = (\beta_{ij})$, $\pi = (\pi_{ij})$, and $\phi = (\phi_k)$ respectively represent dual multipliers associated with constraints (2.1), the two constraint sets (left- and right-hand respectively) in (2.2), the constraints in (2.3), and (2.5), and the constraints in X_1, the Lagrangian dual we formulate is:

$$\text{maximize} \quad \{f(u,\alpha,v,\beta,\pi,\phi): \alpha, v, \beta, \pi \geqslant 0, u \text{ and } \phi \text{ unrestricted in sign}\} \tag{LD}$$

$$\text{where} \quad f(u,\alpha,v,\beta,\pi,\phi) \equiv T + \text{minimum}\{\sum_i \tilde{c}_i x_i + \sum_{i<j}\sum \tilde{D}_{ij} w_{ij}\}$$

$$\text{subject to} \quad 0 \leqslant w_{ij} \leqslant x_j \quad \forall (i,j) \quad \text{with} \quad i<j, \tag{3.10}$$

$$v_l \leqslant \sum x_i \leqslant v_u, \quad 0 \leqslant x_i \leqslant 1 \quad \forall i.$$

The quantities T, $\tilde{c}_i$, and $\tilde{D}_{ij}$ are linear functions of u, α, v, β, π and ϕ obtained by appropriately grouping terms after dualizing (2.1), (2.2), (2.3), (2.5), and the $x \in X_1$

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constraints. Note that the constraints in the subproblem (3.10) are (2.4), (2.6), (3.1), and $x \in X_3$. Furthermore, one may rewrite (3.10) as

$$f(u,\alpha,v,\beta,\pi,\phi) \equiv T + \text{minimum } \{\sum_i [\tilde{c}_i + \sum_{j<i} \min\{0, \tilde{D}_{ji}\}]x_i : (3.1) \text{ and } x \in X_3\}.$$

Problem LD was solved using the subgradient optimization scheme of Poljak (1967), as modified by Held, Wolfe and Crowder (1974) and by Bazaraa and Sherali (1981). Specifically, the step size selection strategy found most effective by Bazaraa and Sherali (1981) was employed with the addition of a line search pattern step (see Bazaraa and Shetty 1979) which duplicates the overall previous step size in the same direction in case an improved incumbent results. For the sake of brevity, we refer the interested reader to Bazaraa and Sherali (1981) for implementation details of this subgradient optimization strategy.

Let $(\bar{u}, \bar{\alpha}, \bar{v}, \bar{\beta}, \bar{\pi}, \bar{\phi})$ be the dual solution upon termination. We terminated the subgradient optimization either when the norm of the subgradient was within a zero tolerance at some iteration or when 250 iterations had elapsed. Now, formulating (3.10) using this dual solution, the optimal dual variables $\bar{\lambda}_{ij}$ associated with the constraints $x_j - w_{ij} \geq 0$ are readily seen to be $\bar{\lambda}_{ij} = -\min\{0, \tilde{D}_{ij}\}\ \forall (i,j)$ with $i < j$. The dual vector $(\bar{u}, \bar{\alpha}, \bar{v}, \bar{\beta}, \bar{\lambda}, \bar{\pi})$ gives the cut (3.2), with $\delta^0 \equiv T$ and with $\gamma_i^0 \equiv \tilde{c}_i - \sum_{j<i} \bar{\lambda}_{ji}\ \forall i$.

3.2. *Updating the Strongest Surrogate Constraint*

In this subsection, we describe the manner in which the strongest surrogate Benders' cut (3.2) is updated in the implicit enumeration scheme described above. To accomplish this, we note that for a given partial solution, defined as before via the index sets I^+, I^-, and I^f at a given node N, Problem MIP reduces to Problem MIP(N), say, as follows. First, if any $i \in I^-$, i.e., x_i is fixed at zero, then $w_{ij} = 0\ \forall j > i$ and $w_{ji} = 0\ \forall j < i$. Second, if any $i \in I^+$, i.e., x_i is fixed at one, then $w_{ij} = x_j\ \forall j > i$ and $w_{ji} = x_j\ \forall j < i$. In this latter case, if for some $i < j$, $x_i = x_j = 1$, then $w_{ij} = 1$ is also implied. Making these substitutions in the objective function produces a constant of value

$$T_1 = \sum_{i \in I^+} c_i + \sum_{i \in I^+} \sum_{\substack{j \in I^+ \\ i<j}} D_{ij}, \tag{3.11}$$

and results in a coefficient of x_i in the objective function, for $i \in I^f$, given by

$$\sigma_i \equiv c_i + \sum_{\substack{j \in I^+ \\ j>i}} D_{ij} + \sum_{\substack{j \in I^+ \\ j<i}} D_{ji} \qquad \forall\, i \in I^f. \tag{3.12}$$

In a similar manner, the constraints (2.1) and (2.2) need only be written for the indices $j \in I^f$, since for $j \in I^- \cup I^+$, these constraints are implied by $x \in X_1 \cap X_2$ and by the fact that the w-variables reduce to the x-variables as described above. Here, $X_2 \subseteq X$ is still implied by the reduced constraint set (2.2), given that $I^f \neq \varnothing$. Let $(2.1)_N$ and $(2.2)_N$ respectively denote (2.1) $\forall$ (j, k) such that $j \in I^f$, and (2.2) $\forall$ (j, l) such that $j \in I^f$, after appropriately substituting for the w_{ij} variables in terms of either constants or x-variables for all pairs of indices (i, j) such that i or $j \in I^f$.

To further reduce the node N problem, define

$$S_N = \{l = 1, \ldots, L : \underset{\substack{0 \leq x_i \leq 1 \\ i \in I^f}}{\text{minimum}} \sum_{i \in I^f} G_{li} x_i < g_l - \sum_{i \in I^+} G_{li}\}. \tag{3.13}$$

Note that S_N denotes the index set of those constraints in X_2 which are not implied by X_3 in the reduced problem. Clearly, the constraints in $(2.2)_N$ indexed by (j, l) with $l \notin S_N$ are redundant even in the continuous sense for Problem MIP(N) and may

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therefore be eliminated from consideration. Let X^N denote the set of constraints in $X_1 \cap X_3$, with x_i fixed appropriately for $i \in I^+ \cup I^-$. Then the reduced program MIP(N) may be stated as follows:

$$\text{minimize} \quad T_1 + \sum_{i \in I^f} \sigma_i x_i + \sum_{i \in I^f} \sum_{\substack{j \in I^f \\ i<j}} D_{ij} w_{ij} \qquad \text{(MIP}(N)\text{)}$$

subject to

$$(2.1)_N, \tag{3.14}$$

$$(2.2)_N \qquad \forall\, l \in S_N, \tag{3.15}$$

$$x_i - w_{ij} \geqslant 0 \qquad \forall\,(i,j) \in I^f, \qquad i<j, \tag{3.16}$$

$$x_j - w_{ij} \geqslant 0 \qquad \forall\,(i,j) \in I^f, \qquad i<j, \tag{3.17}$$

$$-x_i - x_j + w_{ij} \geqslant -1 \qquad \forall\,(i,j) \in I^f, \qquad i<j, \tag{3.18}$$

$$w_{ij} \geqslant 0, \qquad \forall\,(i,j) \in I^f, \qquad i<j, \qquad x \in X^N, \qquad x \text{ binary}. \tag{3.19}$$

In obtaining the reduced program MIP(N) we are, in effect, for a given node N, exploiting the fact that certain dual multipliers with respect to $\overline{\text{MIP}}$ may be expressed (at optimality) in terms of the remaining dual multipliers. Of course, for those constraints that are redundant, a further reduction of the dual space can be accomplished through the assignment of zero-valued optimal dual multipliers. Those multipliers which may be eliminated from consideration, either through substitution or through the assignment of constant (zero) values, are precisely those associated with the constraints in MIP which do not appear in MIP(N). Thus, by considering only the constraints (3.14) through (3.19), we may update (3.2) in the same manner as if the entire constraint set of Problem MIP was considered.

From the viewpoint of deriving strong bounds, the ideal way of updating (3.2) to obtain (3.4) is to determine the optimal dual multipliers associated with the remaining constraints (3.14) through (3.18) of $\overline{\text{MIP}(N)}$, the continuous relaxation of MIP(N), and let (3.4) be the resulting strongest surrogate constraint. On the other hand, one can simply use (3.2) itself by defining the values in (3.4) to be $\gamma_i^N \equiv \gamma_i^0$ $\forall$ $i \in I^f$ and $\delta^N \equiv \delta^0 + \sum_{i \in I^+} \gamma_i^0$. Although the first alternative yields the stronger cut, it is computationally expensive. The second alternative, however, may yield a weak cut, but it is computationally inexpensive. The strategy we adopt compromises between these two alternatives.

Suppose that for the constraints (3.14) through (3.18) we let the dual multipliers be those derived for obtaining (3.2) in the foregoing subsection. That is, suppose we associate $\bar{u}_{jk}$ $\forall$ (j, k) such that $j \in I^f$ with (3.14); $\bar{\alpha}_{jl}$ and $\bar{v}_{jl}$ $\forall\,(j,l)$ such that $j \in I^f$ and $l \in S_N$ with the two constraint sets (left- and right-hand respectively) in (3.15); $\bar{\beta}_{ij}$, $\forall$ $(i, j) \in I^f$, $i < j$, with (3.16); $\bar{\lambda}_{ij}$, $\forall$ $(i, j) \in I^f$, $i < j$, with (3.17); and $\bar{\pi}_{ij}$, $\forall$ $(i, j) \in I^f$, $i < j$, with (3.18). Then, we might no longer have dual feasibility with respect to the w_{ij} columns. If we could restore this dual feasibility, then these multipliers could be used to obtain a valid cut.

This dual feasibility may be restored in several ways. Suppose we look at the reduced costs R_{ij} of the w_{ij} variables in $\overline{\text{MIP}(N)}$ after dualizing (3.14) through (3.18) using the above multipliers. If all these reduced costs are nonnegative, then no revision is necessary for obtaining dual feasibility. Dual feasibility with respect to the x-variable columns can be obtained subsequently by adjusting the multipliers for the constraints $x \in X^N$. If, however, some w_{ij} has $R_{ij} < 0$, then one could increase the dual multipliers associated with both $x_i - w_{ij} \geqslant 0$ in (3.16), and $x_j - w_{ij} \geqslant 0$ in (3.17) by $-R_{ij}/2$ to compensate. (This would also decrease the reduced cost coefficients of x_i and x_j by the same amount.) Alternatively, one could use some redundant constraint added to the

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problem for the specific purpose of permitting one to adjust these multipliers. For example, assume that the redundant constraints $-w_{ij} \geqslant -1 \ \forall \ (i, j) \in I^f, i < j$, have been added to $\overline{\text{MIP}(N)}$. Then, by letting the dual multipliers with respect to these new constraints be $-\min \{0, R_{ij}\} \ \forall \ (i, j) \in I^f, i < j$, dual feasibility may be achieved at the expense of adding the negative R_{ij} coefficients to the cut constant.

The first approach is preferable since it affects the coefficients of the variables rather than the constant term, which always figures into the cut expression. The cut (3.4) may then be verified to have the constant term given by

$$\delta^N \equiv T_1 + T_2 \tag{3.20}$$

where T_1 is defined in (3.11) and where T_2 is the sum of the dualized constants in the constraints and is given by

$$T_2 = \sum_{j \in I^f} \sum_{l \in S_N} [g_l - \sum_{i \in I^+} G_{li}] \bar{\alpha}_{jl} - \sum_{i \in I^f} \sum_{\substack{j \in I^f \\ i<j}} \bar{\pi}_{ij}. \tag{3.21}$$

Similarly,

$$\gamma_i^N = \sigma_i + \rho_i \qquad \forall \, i \in I^f, \tag{3.22}$$

where σ_i is given by (3.12) for $i \in I^J$ and where ρ_i, for $i \in I^J$, is computed by the formulae given below. (Note that the forms of the quantities given here are indicative of the manner in which information may be retained to facilitate the updating of (3.4) from one partial solution to another.) Compute:

$$\rho_i = \sum_{j \in I^f} b_{ij} \qquad \forall \, i \in I^f \qquad \text{where} \tag{3.23}$$

$$b_{ii} = -\sum_k \bar{u}_{ik}(a_{ki} - b_k + \sum_{t \in I^+} a_{kt}) - \sum_{l \in S_N} \bar{v}_{il}(G_{li} - g_l + \sum_{t \in I^+} G_{lt})$$
$$- \sum_{l \in S_N} \bar{\alpha}_{il}(g_l - \sum_{t \in I^+} G_{lt}) \qquad \forall \, i \in I^f \qquad \text{and where} \tag{3.24a}$$

$$b_{ij} = \begin{cases} -\sum_{l \in S_N} \bar{\alpha}_{jl} G_{li} - \bar{\beta}_{ij} + \bar{\pi}_{ij} + \frac{1}{2} \min \{0, R_{ij}\} & \forall \, j > i, \quad j \in I^f \\ -\sum_{l \in S_N} \bar{\alpha}_{jl} G_{li} - \bar{\lambda}_{ji} + \bar{\pi}_{ji} + \frac{1}{2} \min \{0, R_{ji}\} & \forall \, j < i, \quad j \in I^f \end{cases} \qquad \text{for each } i \in I^f. \tag{3.24b}$$

Hence, (3.4) may be updated from one partial solution to another (rather than be recomputed) by using the above formulae. Note that the quantities $\sum_{t \in I^+} a_{kt} \ \forall \ k$ and $\sum_{t \in I^+} G_{lt} \ \forall \ l \in S_N$, as well as the set S_N, are readily available as a result of the logical tests conducted on the set X. Hence, these quantities may be stored to facilitate in the derivation of (3.4).

3.3. *Generation of Benders' Cuts*

In this subsection, we prescribe the manner in which a Benders' (1962) cut of the form (3.6), corresponding to a binary vector $x = \bar{x}$ is generated. Given a fixed $x = \bar{x}$, a Benders' cut may be obtained by dualizing constraints (2.1) through (2.5) of Problem MIP using an optimal set of dual multipliers for that linear program over the w-variables obtained by inserting $x = \bar{x}$ into Problem MIP. As pointed out by Magnanti and Wong (1981), the strength of the Benders' cut obtained may depend on the choice of the dual optimal solution, in case alternate optimal dual solutions exist. In Theorem 3 below, we present in closed form optimal dual multipliers for constraints (2.1) through (2.5) of Problem MIP, given that x is fixed at the binary vector $\bar{x}$. Thereafter,

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we explain the strategy used for selecting from amongst the multipliers that dual solution from which a Benders' cut is generated.

THEOREM 3. *Consider the linear program over the w-variables obtained by fixing $x \equiv \tilde{x}$, a binary vector, in Problem MIP. Let (u_{jk}), (α_{jl}), (v_{jl}), (β_{ij}), (λ_{ij}), and (π_{ij}) denote the dual multipliers associated with constraints* (2.1), *the two constraint sets (left- and right-hand respectively) in* (2.2), *and the constraint sets* (2.3), (2.4), *and* (2.5) *respectively. Then $(\tilde{u}, \tilde{\alpha}, \tilde{v}, \tilde{\beta}, \tilde{\lambda}, \tilde{\pi})$ defined below yields an optimal set of dual multipliers for this linear program:*

$$\tilde{u}_{jk}=0 \qquad \forall (j,k), \qquad \tilde{\alpha}_{jl}=0 \qquad \forall (j,l), \qquad \tilde{v}_{jl}=0 \qquad \forall (j,l),$$

$$\tilde{\beta}_{ij}=\begin{cases} -D_{ij}[\theta_{ij}(1-\tilde{x}_i)+(1-\theta_{ij})\tilde{x}_j] & \text{if } D_{ij}<0 \\ 0 & \text{otherwise,} \end{cases} \qquad \forall (i,j), \quad i<j, \tag{3.25}$$

$$\tilde{\lambda}_{ij}=\begin{cases} -D_{ij}[\theta_{ij}\tilde{x}_i+(1-\theta_{ij})(1-\tilde{x}_j)] & \text{if } D_{ij}<0 \\ 0 & \text{otherwise,} \end{cases} \qquad \forall (i,j), \quad i<j, \tag{3.26}$$

$$\tilde{\pi}_{ij}=\begin{cases} D_{ij}\theta_{ij} & \text{if } D_{ij}\geqslant 0 \\ 0 & \text{otherwise.} \end{cases} \qquad \forall (i,j), \quad i<j, \tag{3.27}$$

Here $0 \leqslant \theta_{ij} \leqslant 1$ are scalars for all (i, j), $i < j$, with $\theta_{ij} = 1$ for all (i, j), $i < j$, such that $D_{ij} > 0$ and $\tilde{x}_i = \tilde{x}_j = 1$, and with $\theta_{ij} = 0$ for all (i, j), $i < j$, such that $D_{ij} > 0$ and $\tilde{x}_i = \tilde{x}_j = 1$.

PROOF. Note that the prescribed dual variables are nonnegative. In order to verify dual feasibility, we need to verify that

$$-\tilde{\beta}_{ij}-\tilde{\lambda}_{ij}+\tilde{\pi}_{ij}\leqslant D_{ij} \qquad \forall (i,j), \qquad i<j. \tag{3.28}$$

For each (i, j) pair with $i < j$, if $D_{ij} < 0$, then the left-hand side of (3.28) is

$$D_{ij}[\theta_{ij}(1-\tilde{x}_i)+(1-\theta_{ij})\tilde{x}_j+\theta_{ij}\tilde{x}_i+(1-\theta_{ij})(1-\tilde{x}_j)]=D_{ij},$$

and if $D_{ij} \geqslant 0$, then the left-hand side of (3.28) is $D_{ij}\theta_{ij} \leqslant D_{ij}$. Hence, the prescribed solution is dual feasible. Its dual objective value is

$$\sum_i c_i\tilde{x}_i-\sum_{i<j}\sum \tilde{x}_i\tilde{\beta}_{ij}-\sum_{i<j}\sum \tilde{x}_j\tilde{\lambda}_{ij}+\sum_{i<j}\sum \tilde{\pi}_{ij}(\tilde{x}_i+\tilde{x}_j-1)$$

$$=\sum_i c_i\tilde{x}_i+\sum_{\substack{i<j\\ D_{ij}<0}}\sum D_{ij}[\theta_{ij}\tilde{x}_i(1-\tilde{x}_i)+(1-\theta_{ij})\tilde{x}_i\tilde{x}_j+\theta_{ij}\tilde{x}_i\tilde{x}_j+(1-\theta_{ij})\tilde{x}_j(1-\tilde{x}_j)]$$

$$+\sum_{\substack{i<j\\ D_{ij}\geqslant 0}}\sum D_{ij}\theta_{ij}[\tilde{x}_i+\tilde{x}_j-1]=\sum_i c_i\tilde{x}_i+\sum_{i<j}\sum \tilde{x}_i\tilde{x}_jD_{ij}\equiv\sum_i c_i\tilde{x}_i+\sum_{\substack{i<j\\ \tilde{x}_i=\tilde{x}_j=1}}\sum D_{ij}$$

which is the primal objective value of MIP with $x \equiv \tilde{x}$ fixed. This completes the proof. □

Observe that the Benders' cut derived from this solution is

$$z\geqslant\sum_i c_ix_i-\sum_{i<j}\sum x_i\tilde{\beta}_{ij}-\sum_{i<j}\sum x_j\tilde{\lambda}_{ij}+\sum_{i<j}\sum \tilde{\pi}_{ij}(x_i+x_j-1) \quad \text{or} \tag{3.29a}$$

$$z\geqslant\sum_i[c_i+\sum_{j>i}\tilde{\pi}_{ij}+\sum_{j<i}\tilde{\pi}_{ji}-\sum_{j>i}\tilde{\beta}_{ij}-\sum_{j<i}\tilde{\lambda}_{ji}]x_i-\sum_{i<j}\sum\tilde{\pi}_{ij}. \tag{3.29b}$$

Recall from the beginning of this section that each Benders' cut (3.6) is computed within the framework of an implicit enumeration scheme at a given node N defined by I^+, I^-, and I^f representing, respectively, the index sets of x-variables fixed at one,

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fixed at zero, and free. Further recall that each Benders' cut generated from some binary vector $\tilde{x}$ is used for computing bounds and conducting logical tests on only that partial solution vector from which it was generated. Thus, in selecting from amongst the alternate optimal dual solutions for deriving a strong cut from the viewpoint of bound computations, we set

$$\theta_{ij}=\begin{cases} 1 & \text{if } i\in I^+, \quad j\in I^f, \quad \text{and} \quad \tilde{x}_j=0 \\ 1 & \text{if } i\in I^f, \quad j\in I^+, \quad \text{and} \quad \tilde{x}_i=0 \\ 0 & \text{if } i\in I^-, \quad j\in I^f, \quad \text{and} \quad \tilde{x}_j=1 \\ 0 & \text{if } i\in I^f, \quad j\in I^-, \quad \text{and} \quad \tilde{x}_i=1 \end{cases} \forall (i,j), \quad i<j, \quad \text{such that} \quad D_{ij}>0, \quad \text{and}$$

$$\theta_{ij}=\begin{cases} 1 & \text{if } i\in I^+, \quad j\in I^f, \quad \text{and} \quad \tilde{x}_i=1 \\ 1 & \text{if } i\in I^-, \quad j\in I^f, \quad \text{and} \quad \tilde{x}_j=0 \\ 0 & \text{if } i\in I^f, \quad j\in I^+, \quad \text{and} \quad \tilde{x}_i=1 \\ 0 & \text{if } i\in I^f, \quad j\in I^-, \quad \text{and} \quad \tilde{x}_i=0. \end{cases} \forall (i,j), \quad i<j, \quad \text{such that} \quad D_{ij}<0$$

For the index pairs $(i, j) \in I^f$, $i < j$, with $D_{ij} > 0$ and $\tilde{x}_i + \tilde{x}_j = 1$, we set $\theta_{ij} = 1$ while for the index pairs $(i, j) \in I^f$, $i < j$, with $D_{ij} < 0$ and either (i) $\tilde{x}_i + \tilde{x}_j = 0$ or (ii) $\tilde{x}_i + \tilde{x}_j = 2$, we set $\theta_{ij} = \frac{1}{2}$, in order to evenly divide the penalty of D_{ij} between the coefficients of the variables x_i and x_j in the cut. For the remaining index pairs (i, j), $i < j$, the Benders' cut is unaffected by the value of θ_{ij} and hence θ_{ij} may be considered to be arbitrarily fixed in the interval [0, 1]. We note that the parameters θ_{ij} defined as above for $(i, j) \in I^f$ may be suitably adjusted to accommodate the specific set X.

4. Computational Experience

In this section, we provide computational experience for solving Problem QP via Problem MIP. As a point of illustration, we chose to compare this performance with that of solving QP via the linearization MIP1 for the following reasons. First, MIP1 is relevant because the methods of Glover and Woolsey (1974), Hammer *et al.* (1984), Peterson (1971) and Rhys (1970) are all directly related to this linearization as pointed out in §2. Second, the linearization MIP is obtained from MIP1 by adding the special constraint sets (2.1) and (2.2), and so it is of interest to demonstrate how these additional constraints which tighten the continuous relaxation help in improving the computational performance of the algorithm. Third, MIP1 is amenable to the same algorithmic strategies as is MIP, and therefore, the differences observed in their relative performances can be directly attributed to the contribution made by the additional constraint sets in MIP. As far as the linearization MIP2 is concerned, we showed in §2 that MIP provides a tighter continuous relaxation in the sense of Theorem 2, and illustrated this with an example. Furthermore, McBride and Yormark (1980) point out that while MIP2 is a more compact representation of QP, it provides a rather weak continuous relaxation, and results in a slow convergence rate. In view of this evidence, and because the development of a specialized algorithm for solving MIP2 is beyond the scope of this paper, we do not provide comparative results using MIP2.

We chose problems with the set X represented by set covering constraints for two reasons. First, constraints of this kind arise in an important application, the dynamic set covering problem, which seeks a sequence of facility relocations over time (see Chrissis *et al.* 1978). Second, as pointed out in §2, this choice lends only moderate additional strength to the linearization, and hence the results would be indicative of what one might expect in no better than an average situation. Note that for a set covering constraint set X, the constraints (2.1) do not appear in Problem MIP and further, for each index pair (j, l) such that $G_{lj} = 1$, the right-hand inequality of (2.2) is implied by (2.6). In addition, the right-hand inequality in (3.1) may be trivially

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omitted. Thus, the results presented here exhibit in a relatively modest fashion the strength of the continuous relaxation, Problem $\overline{\text{MIP}}$, of Problem MIP. Nonetheless, since only a limited number of test problems were run, all with the same set covering types of constraints, some caution must be exercised in inferring the general superiority of the proposed approach.

Tables 4.1 and 4.2 present results for a set of test problems. The runs were designed to demonstrate the relative performances of Problems MIP and MIP1 as a function of the problem size and the density of X. In Table 4.1, the objective coefficients D_{ij} were generated as $D_{ij} = f_{ij} + f_{ji}\ \forall\ (i, j),\ i < j$, where the f_{ij} coefficients were generated uniformly on the interval [0, 25], $\forall\ (i, j)$. In order to ensure that the quadratic terms dominate the linear terms, the c_i coefficients were generated uniformly on the interval $[-2.5(m-1), 0]\ \forall\ i$. The columns in Tables 4.1 and 4.2 can be interpreted as follows. Nine problem runs are presented, each problem being solved using both MIP and MIP1 as indicated in the first column. The first six problems have a density of 0.1 for the set X, and the last three have a density of 0.25. (After generating X, each row was checked to be sure that it had at least two nonzero entries. No special set covering logical tests were used.) The column X gives the number of rows and variables in the set covering constraint set X; "ν(QP)" gives the optimal integer value; "$\nu(\overline{\text{MIP}})$ or $\nu(\overline{\text{MIP1}})$" gives the linear programming objective value of the dual to $\overline{\text{MIP}}$ or $\overline{\text{MIP1}}$, as appropriate, upon termination of the subgradient optimization scheme; "n_N" denotes the number of nodes enumerated in the implicit enumeration procedure;

TABLE 4.1

Positive D_{ij} and Negative c_i Coefficient Problems

Problem	X	ν(QP)	$\nu(\overline{\text{MIP}})$ or $\nu(\overline{\text{MIP1}})$	n_N	n_E	r_F	T_{LP}	T_T
1MIP	(10, 30)	159	157.92	16	0	$\frac{5}{12}$	5.43	5.51
1MIP1			−441.60	3154	4	$\frac{1058}{1128}$	1.12	7.11
2MIP	(10, 50)	−258	−490.72	1068	11	$\frac{202}{645}$	14.39	21.43
2MIP1			−1457.36	22572	13	$\frac{1998}{3875}$	2.96	68.27
3MIP	(10, 70)	−669	−1708.10	37838	27	$\frac{509}{11982}$	27.49	290.28
3MIP1			−3289.25	172746	61	$\frac{665}{16054}$	5.92	821.17
4MIP	(25, 30)	556	553.08	16	0	$\frac{5}{12}$	9.60	9.79
4MIP1			−510.44	14388	12	$\frac{2230}{2273}$	1.25	24.53
5MIP	(25, 50)	29	−23.98	106	0	$\frac{45}{95}$	25.81	27.71
*5MIP1			−1761.60	>383000	—	—	3.31	>1200
6MIP	(25, 70)	−258	−375.11	366	5	$\frac{165}{345}$	52.40	62.77
*6MIP1			−2710.10	>214000	—	—	6.27	>1200
7MIP	(25, 30)	11	−17.99	60	0	$\frac{21}{48}$	11.00	11.58
7MIP1			−530.12	2008	11	$\frac{294}{401}$	1.31	6.31
8MIP	(25, 50)	−317	−800.05	5170	14	$\frac{280}{1515}$	29.89	64.85
8MIP1			−1588.05	22674	36	$\frac{901}{2718}$	3.33	96.58
9MIP	(25, 70)	−631	−1981.55	40794	32	$\frac{460}{8186}$	55.35	403.30
*9MIP1			−2850.35	>160000	—	—	6.10	>1200

* Failed to solve the problem in a 1200 cpu second time limit.

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TABLE 4.2

Positive D_{ij} and c_i Coefficient Problems

Problem	X	ν(QP)	$\nu(\overline{\text{MIP}})$ or $\nu(\overline{\text{MIP1}})$	n_N	n_E	r_F	T_{LP}	T_T
1MIP	(40, 30)	2155	2141.07	24	0	$\frac{8}{19}$	15.26	15.91
1MIP1			411.13	564	0	$\frac{233}{292}$	1.51	5.36
2MIP	(40, 50)	3956	3890.10	50	0	$\frac{15}{39}$	40.77	42.20
*2MIP1			993.19	>52000	—	—	3.69	>1200
3MIP	(40, 70)	3559	3547.47	34	0	$\frac{14}{30}$	75.33	76.51
*3MIP1			757.68	>94000	—	—	6.54	>1200
4MIP	(40, 30)	1265	1265	0	0	—	12.26	12.33
4MIP1			313.90	1210	0	$\frac{596}{612}$	1.398	12.38
5MIP	(40, 50)	1833	1757.61	54	0	$\frac{25}{52}$	40.11	43.23
5MIP1			385.35	61390	0	$\frac{26045}{26989}$	3.64	608.55
6MIP	(40, 70)	1312	1288.73	14	0	$\frac{7}{13}$	77.22	78.68
6MIP1			397.80	2130	0	$\frac{1063}{1845}$	6.54	196.79

* Failed to solve the problem in a 1200 cpu second time limit.

"n_E" gives the number of times the incumbent was (further) improved by virtue of the local explorations at each new incumbent; "r_F" gives a ratio p/q, with $q \equiv$ the number of times the linear program (3.9) was solved for bound computations, and with $p \equiv$ the number of times this bound fathomed the corresponding node; "T_{LP}" denotes the time in cpu seconds to obtain the strongest surrogate constraint (3.2); and "T_T" gives the total execution time in cpu seconds to solve Problem QP. The coding was done in Fortran IV and all problems were run in batch mode at roughly the same load on an IBM 3081 Series D24 Group K computer, using the Fortran H extended compiler on optimization level 2.

In all cases, the time for the implicit enumeration phase (beyond the linear programming effort to obtain (3.2)) was considerably less for MIP than for MIP1. The number of nodes enumerated (n_N) gives the same indication. Moreover, for the first six (lower density) problems, the total execution time for solving Problem QP via Problem MIP was significantly smaller than the total execution time when Problem MIP1 was used, with the difference increasing dramatically with problem size. Note that most real-world problems do have densities less than 0.1. For the high density problems (7, 8, and 9), by the nature of the set covering constraints, the additional strength lent to MIP over MIP1 by constraints (2.1) and (2.2) is somewhat smaller, and the *overall* benefits derived from MIP over MIP1 do not manifest themselves until the problem size increases sufficiently, although the total number of nodes enumerated is always significantly smaller for MIP.

We also attempted to evaluate the benefits of the other strategies employed in the algorithm. For example, when the Benders' cuts of §3.3 were not used in the algorithm in coordination with the strongest surrogate Benders' cut, then while problem runs 1MIP, 1MIP1, and 2MIP remained almost unaffected, the runs 2MIP1, 3MIP, and 3MIP1 did not terminate within a 1200 cpu second time limit. Furthermore, the local exploration step proved worthwhile because of the frequency with which improved incumbent solutions were discovered. These local explorations helped in reducing the

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sensitivity of the algorithm with respect to whether a breadth-first or depth-first approach was adopted. (We made our runs with a combination of the two approaches, permitting a maximum of two active nodes.) Also, the linear programming bound (3.9) enhanced the performance when computed at judiciously chosen nodes as indicated in §3. In general, the tighter the linearization, the more beneficial was the computation of this bound.

As a matter of interest, we also generated positive cost coefficient problems with the D_{ij} coefficients being generated as for Table 4.1, and with the c_i coefficients generated uniformly on the interval $[0, 2.5(m-1)]$ $\forall$ i. Table 4.2 gives the results. With positive cost coefficients, an optimal solution is a prime cover, and because of its resemblance to set covering problems, these problems turn out to be relatively easier to solve. Consequently, the problem size needs to increase sufficiently before the benefits of the stronger linearization become clear. Since an optimal solution to each of these problems is a prime cover, the local explorations do not assist by discovering new incumbent solutions as a result of the nature of the implicit enumeration scheme. Problems 1, 2, and 3 have a density of 0.05 for the set X and Problems 4, 5, and 6 have a density of 0.1. As before, a smaller density magnifies the relative worth of the stronger linearization, and in any case, the number of nodes generated using MIP are significantly fewer than those generated via MIP1.

5. Conclusion

In this paper, we have presented a new linearization MIP for the linearly constrained zero-one quadratic programming problem. This linearization has been shown to dominate other linearizations in the literature with respect to the strength of the linear programming relaxation. In particular, MIP tightens a certain linearization MIP1, which has been previously proposed in various contexts, by adding several judiciously constructed constraints. An implicit enumeration algorithm using Lagrangian relaxation, Benders' cutting planes and local explorations has been designed in order to exploit the strength of the proposed linearization, without being encumbered by its size. The computational experience reported has provided a strong indication of the benefits of strengthening the linearization MIP1 through the additional constraints appearing in MIP. The idea behind the linearization technique, as well as the algorithmic strategies used, can both be extended to other similar situations, such as the mixed-integer zero-one quadratic programming problem involving cross-product terms between the zero-one variables and between the zero-one and the continuous variables.[1]

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